

Predicting Valorant Match Outcomes

Using historical team performance and match context features

COMP 3608 Project

Binary Classification

VCT 2023-2025 Data

Project Focus

What we are trying to predict and why it matters

Problem

Goal: predict whether Team A wins or loses a Valorant match.

The problem is framed as a binary classification task.

The model uses pre-match information only, so predictions reflect a realistic pre-game setting.

The project focuses on comparing algorithms and explaining their tradeoffs, not only chasing one best score.

Target Variable

1 = Team A Wins

0 = Team A Loses

Models can also output win probabilities for interpretation, such as a 64% predicted chance of winning.

Stakeholders and Value

Who can use the predictions?



Coaches

Prepare for high-risk matchups and target weaknesses



Analysts

Understand which indicators are most predictive



Team Managers

Support scouting, roster, and strategic decisions



Commentators

Create data-supported pre-match narratives



Organizers

Identify competitive matchups for scheduling and coverage

Data Used

Three yearly Kaggle data subsets from Valorant Champions Tour

330

2023 matches

434

2024 matches

501

2025 matches

1265

total matches

Dataset Scope

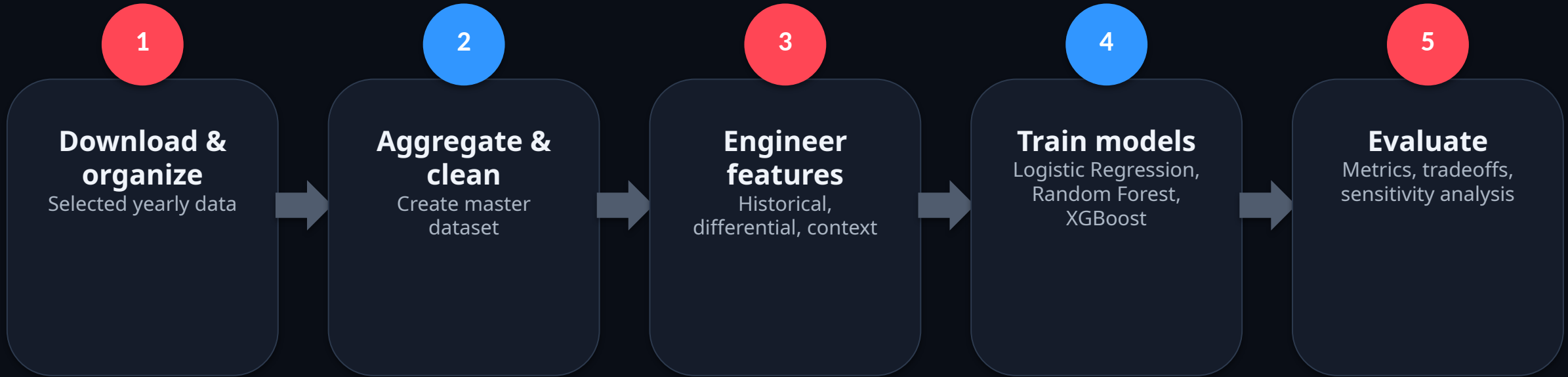
The project uses the 2023, 2024, and 2025 yearly subsets from the VCT Kaggle data.

The data was downloaded, aggregated, cleaned, and transformed into a master modelling dataset.

Using recent seasons keeps the project focused on more current competitive trends and reduces unnecessary complexity.

Project Pipeline

From raw data to model comparison



The notebook contains the full technical detail; the write up summarizes the project according to the rubric.

Feature Engineering

Turning raw match data into predictive signals

Historical Form

Team win rate

Average rating

Map win percentage

Differential Strength

Win rate difference

Rating difference

Map win percentage
difference

Match Context

Stage stakes

Elimination match

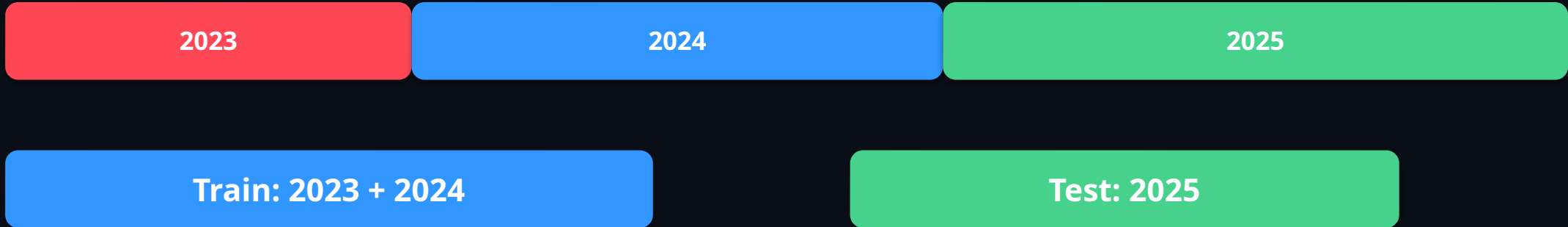
Grand final flag

Draft / ban first

Main idea: relative team strength is more useful than raw statistics alone.

Experimental Design

A time-aware setup to avoid future data leakage



Why this matters

Older matches are used for training and newer matches are held out for testing.

Walk-forward validation checks whether models generalize from one season into the next.

Algorithms Compared

Three approaches plus a simple baseline

Majority Baseline

Performance floor; predicts the most common class only

Logistic Regression

Interpretable model for probability and feature direction

Random Forest

Nonlinear ensemble model that captures feature interactions

XGBoost

Boosted tree model designed for stronger tabular performance

Evaluation Metrics

How model performance was judged

Accuracy

Overall correctness

Precision

How reliable predicted wins are

Recall

How many actual wins are captured

F1 Score

Balance of precision and recall

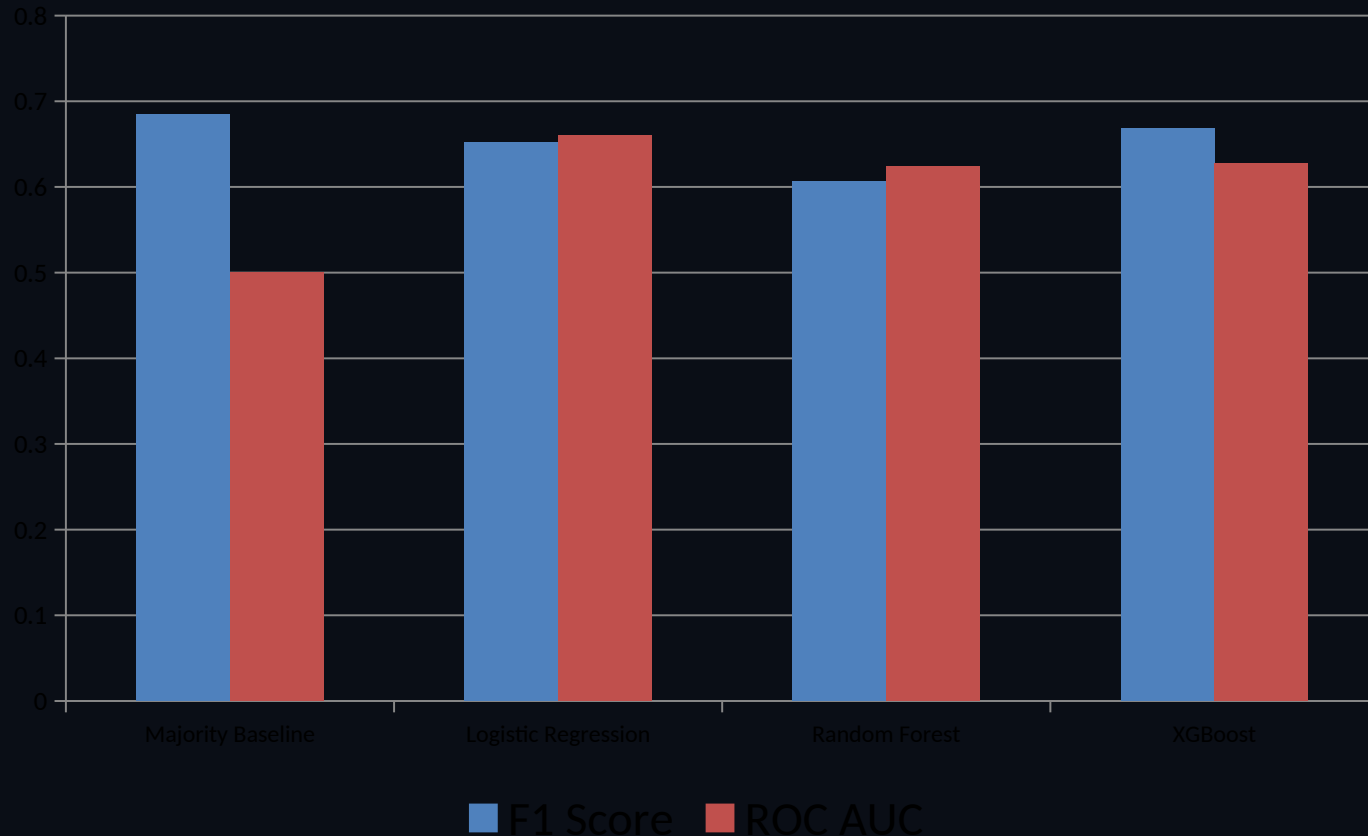
ROC AUC

Ability to separate wins from losses

ROC AUC is especially useful because the majority baseline can have a high F1 score while still learning nothing useful.

Final 2025 Test Results

Model performance on the held-out season



Main takeaways

Logistic Regression had the best ROC AUC, meaning it separated likely winners from losers best.

XGBoost had the best F1 score among learned models.

The baseline F1 is misleading because it predicts Team A wins every match.

Interpretation and Tradeoffs

No model is perfect, so the best choice depends on the use case

Logistic Regression

Best for explanation

Strong ROC AUC and easy to interpret, but may miss nonlinear patterns

Random Forest

Middle ground

Handles nonlinear relationships, but performed weaker in this experiment

XGBoost

Best raw F1

Stronger classification performance, but less transparent and needs careful tuning

Best stakeholder-friendly option: Logistic Regression. Best predictive option by F1: XGBoost.

Limitations and Future Improvements

What the model does not fully capture

Limitations

Season-to-season changes can affect performance: roster changes, meta shifts, coaching changes.

New or less active teams have limited history, creating a cold-start problem.

The model predicts match outcomes, not individual map outcomes or in-series adjustments.

Future Work

Add map-specific team performance and opponent strength ratings.

Include player-level form, agent composition, and roster change indicators.

Explore Elo-style regional ratings or more detailed round-by-round momentum features.

Conclusion

What the project showed

Final message

The project successfully framed VCT match prediction as a binary classification problem.

Historical team strength and differential features provided useful predictive signal.

Logistic Regression was the strongest for ROC AUC and interpretation.

XGBoost produced the strongest F1 score among learned models.

The results show that pre-match historical data can help predict outcomes, but esports matches remain uncertain and high variance.

Thank you